HACKATHON THEME · BEST USE OF n8n — HEALTHCARE AUTOMATION

MedConnect Pulse

Connecting Care When Seconds Count

THE PROBLEM SPARK · **Born from the 2021 Oxygen Crisis**

Hospitals operate as isolated islands: front desks drown in repetitive calls, patients wait blind for hours, and ER doctors lose golden-hour minutes phoning around for blood, oxygen, and ICU beds.

COLLEGE: JIMS Engineering Management Technical Campus **TEAM:** 4 Bit Minds **TEAM LEADER:** Ayank Kumar



IDEA TITLE · PROPOSED SOLUTION

One Platform, Two Connected Pillars



MedConnect Pulse is an n8n-powered, event-driven automation platform that turns isolated hospitals into one connected regional safety net — pairing a multilingual AI receptionist with an inter-hospital emergency supply network.

AI Receptionist & Live Queue Engine

Multi-channel voice intake

Web mic, WhatsApp & Telegram voice notes transcribed by Sarvam AI, understood by an OpenAI agent.

Instant booking tokens

Auto-books OPD slots in Supabase and issues a token like #MED-4029 in seconds.

Live wait-time alerts

n8n recalculates queue position on every check-in and pushes it straight to WhatsApp.

10km Emergency Supply Dispatch

Nearest Hospital Alert Engine

the platform instantly finds hospitals within a 10 km radius and broadcasts emergency resource requests only to the most relevant responders.

1-click accept & lock

Telegram inline buttons let ER managers accept instantly; n8n locks the stock to stop double-allocation.

Auto-escalation + QR pass

No response in 3 minutes triggers a Twilio call; acceptance generates a tamper-proof QR dispatch pass.





Two Journeys, One n8n Engine

From a spoken word to a booked slot. From an emergency alert to a dispatched unit — every step below runs as a single automated n8n workflow.

PATIENT JOURNEY · AI RECEPTIONIST

Speaks Naturally

Web mic, WhatsApp or Telegram voice note — any Indic language.

Sarvam AI Transcribes

saaras:v1 STT converts accented regional speech to clean text.

OpenAI Reads Intent

gpt-4o-mini extracts the request — book, check, or cancel.

Supabase Books Slot

Slot reserved instantly; double-booking is prevented.

Token + Live Alerts

#MED-4029 issued; WhatsApp tracks wait time in real time.

EMERGENCY JOURNEY · GEO-DISPATCH

Doctor Raises Alert

Selects O- blood, ICU bed or oxygen in one tap.

10 km Radius Filter

n8n code node targets hospitals in radius — no Maps API.

Telegram Broadcast

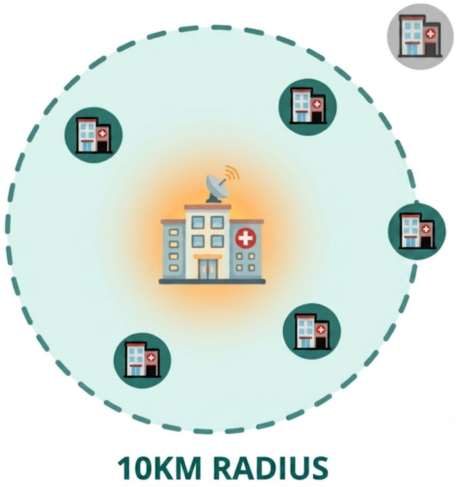
[ACCEPT & SUPPLY] button sent to nearby ER managers.

ACCEPTED · WITHIN 3 MIN

Stock locked instantly; tamper-proof QR dispatch pass generated.

TIMEOUT · NO RESPONSE

Automated Twilio voice call escalates to regional coordinators.



Why It's Different

Four architectural bets that let a 4-person team ship in 36 hours what legacy hospital software can't do at all.



N8N : WORKFLOW

n8n-native

Booking, emergency dispatch, alerts and escalations run inside one automated n8n workflow—no manual coordination required.

SARVAM : VOICE ASSISTANT

Sarvam AI · saaras:v1

Understands Indian languages, converts speech to text, extracts user intent, and responds naturally for appointments and emergency requests.

RAPID RESPONSE ENGINE

Telegram → Twilio fallback

Nearby hospitals receive alerts instantly. If there's no response within 3 minutes, the system automatically escalates to the next available hospital.

GEO-FILTER

No paid Maps API

Finds the nearest suitable hospitals within a configurable radius using real-time location filtering—without relying on paid Maps APIs.

Bonus: India's first platform combining AI Reception, Live Queue Engine ,Emergency Dispatch and Hospital Resource Exchange.

TECHNICAL APPROACH

Tech Stack & Automation Pipeline

n8n
Workflow Engine

Sarvam AI
saaras:v1 STT

OpenAI
gpt-4o-mini

Supabase
Postgres DB

React / Vite
Dashboard

Telegram
Inline Alerts

Twilio
Voice Fallback

Geo-Filter
15km Filter

WF1 · VOICE-POWERED AI RECEPTIONIST

Voice Input
(Web / WhatsApp)
→
Sarvam STT
Node Translation
→
OpenAI Agent
(Intent + Slot)
→
Supabase
Booking Reservation
→
WhatsApp
Token Confirmation

WF2 · EMERGENCY GEO-DISPATCH

Emergency
Webhook Trigger
→
Geo-Filter
10km GPS Filter
→
Telegram Alert
[Accept & Supply]
→
3-Min Wait Node
(Timeout Check)
→
QR Pass / Call
Twilio Escalation

Methodology: Event-driven architecture using n8n workflows, AI intent recognition, geo-based hospital filtering, and automated escalation with real-time notifications.



IMPACT & BENEFITS

Built for the Golden Hour

<45s	80%	10km	10k+
Emergency match & response latency	Routine OPD inquiries fully automated	Regional emergency safety-net radius	Daily workflow runs per automation node



Lives Saved in the Golden Hour

Cuts inter-hospital transfer coordination from 25–40 minutes down to under 30 seconds during shortages.



Zero Queue Chaos

Live Telegram wait-time tracking turns crowded, opaque waiting rooms into transparent, predictable schedules.



Zero-Infrastructure Network

Powered by n8n, Supabase, and cloud services, enabling hospitals to deploy quickly without expensive servers or complex IT maintenance.

THANK YOU

MedConnect Pulse

“Turning isolated hospital silos into one connected, event-driven regional emergency safety net.”



Ayank Kumar

Team Leader · Dev 1

n8n Workflows & Voice AI (Sarvam + OpenAI)



Ishan Varshney

Backend & Logic Lead · Dev 2

Supabase DB, Haversine, Telegram & Twilio



Ishaan Sama

Frontend Engineer · Dev 3

React / Vite 3-in-1 Dashboard & Voice Widget



Bhavya Gupta

Integration & Pitch · Dev 4

E2E Testing, Demo Fail-Safes & Presentation

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